

Roll No.

Total Pages : 4

11062/NJ**D-18/2111****PYTHON PROGRAMMING**

Paper-202

Semester-III

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. Discuss various features of Python. 5

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[P. T. O.]

2. What are shared references? Explain in detail with the help of an example. 5
3. Write a program to create a class showing how to access an attribute and the method contained in the class through an object. 5
4. Explain the following : 5
- (i) Dictionary
- (ii) List Comprehension.
5. Explain the concept of indexing and slicing in numpy arrays with example of each. 5

SECTION—B

6. Differentiate between Series and Data Frame with the help of an example. 5
7. Explain, how the following problems can be handled : 5
- (i) Missing values
- (ii) Duplicate values. 5

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8. Write a program to display a line chart of sine wave using random values showing the attributes such as a label, title and a legend. 5
9. Explain the following terms with reference to Data Frame : 5
- (i) sort_values
 - (ii) sort_index
 - (iii) astype
 - (iv) loc
 - (v) iloc.
10. Explain the creation of a series object from a list with the help of an example. 5

SECTION—C

11. Answer the following questions : 10×2=20
- (i) Explain the concept of Broadcasting.
 - (ii) What are Categorical attributes? Explain with the help of an example.

- (iii) Explain the range in python.
- (iv) Differentiate between Global and Local variables with example.
- (v) Define the Subplot.
- (vi) What is the use of inplace parameter?
- (vii) Write the name of a function that is used to read the csv file in pandas.
- (viii) Name any two charts that can be created in Python.
- (xi) Differentiate between a vector and a matrix.
- (x) Explain about a List and a Tuple.

D-28/2110

6902/NJ

Python Programming (CPE-202)
(Sem-III)

Time Allowed: 3 Hours

Maximum Marks: 50

Note: The candidates are required to attempt **three** questions each from Section A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

Section –A

1. Differentiate between mutable and immutable objects with the help of an example. (5)
2. What are lambda functions? Explain in detail with the help of an example. (5)
3. Write a program to print factorial of a number. (5)
4. What is a List and what different basic functions can be performed on it. (5)
5. Explain the concept of indexing and slicing in numpy arrays with example of each. (5)

Section – B

6. Explain with example how the rows and columns can be extracted from a data frame. (5)
7. (a). Explain figure() and plot() with suitable example. (5)
(b). Give steps for importing an excel file in Pandas.
8. Explain how the rows and columns can be deleted in a data frame. (5)
9. Explain the following terms with reference to Series: (5)
 - a. count
 - b. describe
 - c. idxmin
 - d. idxmax
 - e. value_counts
10. Explain the creation of a series object from a list with the help of an example. (5)

Section – C

11. Answer the following: ($10 \times 2 = 20$)
 - i. What are identifiers.
 - ii. What are Categorical attributes. Explain with the help of an example.
 - iii. Explain the concept of class and an object in brief.
 - iv. Differentiate between global and local variables with example.
 - v. Name any four data types in Python.
 - vi. Explain how a subplot is different from a plot?
 - vii. Explain the concept of a tuple with an example.
 - viii. Name any two charts that can be created in Python.
 - ix. Explain any two series attributes.
 - x. Explain about list comprehension.

SECTION—A

1. Discuss the important features of Python. 5

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[P. T. O.]

2. Explain the following :

(a) Mutable and Immutable objects

(b) Tuples. 5

3. Write a program to Print factorial of a number. 5

4. Explain list creation along with addition and deletion of elements with example. 5

5. Explain the following :

(a) Dictionary

(b) Function.

5

SECTION—B

6. Explain creation of Data frame using list and dictionary.

5

7. Explain the following in data frames :

(a) Identifying missing value.

(b) Deleting row and column.

5

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8. Explain creation of series from a list with example.

5

9. Explain the following terms with reference to series :

(a) Value_counts

(b) Sort_value

(c) Idmax

(d) Describe

(e) Count.

5

10. (a) Explain figure() and plot() with suitable example.

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(b) Give steps in importing excel files in Randa.

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SECTION—C

11. Write short notes on the following : $10 \times 2 = 20$

(i) Define the array with example.

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[P. T. O.]

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(ii) Give the basic input output statements in Python.

(iii) Explain the range in Python.

(iv) Explain Global and Local variable in short.

- (iii) Explain the range in Python.
- (iv) Explain Global and Local variable in short.
- (v) What are Identifiers ?
- (vi) What are modules in Python ?
- (vii) Explain the class with example.
- (viii) Name any four datatypes available in Python.
- (ix) Explain the object in Python.
- (x) Name any two charts which can be created in Python.

G-53/2114

PYTHON PROGRAMMING

Paper-2301T

Semester-III

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. Write various features of Python.

5

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[P. T. O.]

2. Explain for loop with a suitable example. 5
3. Explain various list operations. 5
4. How to create a class and object? How to access the class members using object? 5
5. What is Numpy arrays? Discuss reshaping arrays with examples. 5

SECTION—B

6. What are the features of Pandas? 5
7. How to extract columns from Dataframe? 5
8. Explain various Dataframe methods? 5

8. Explain various Dataframe methods? 5
9. How to create scatter chart and line chart in python visualization? 5
10. Define the term Multi Indexing and Raindexing in pandas. 5

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SECTION—C

11. Answer the following questions : 10×2=20

SECTION—C

11. Answer the following questions :

10×2=20

- (i) What is Python?
- (ii) Define scope of variable.
- (iii) What is shared reference in python?
- (iv) Discuss lambda function.
- (v) What is tuple?
- (vi) What is ndim attribute of pandas?
- (vii) How to delete row from a dataframe?
- (viii) List various series attributes.
- (ix) What is Matplotlib?
- (x) How python is different from other language?